COMPUTING LAB - II
ADVANCED SYSTEM SOFTWARE LAB - CSE 5261
MTech - CSE
Week - 1 Basics of Python Programming

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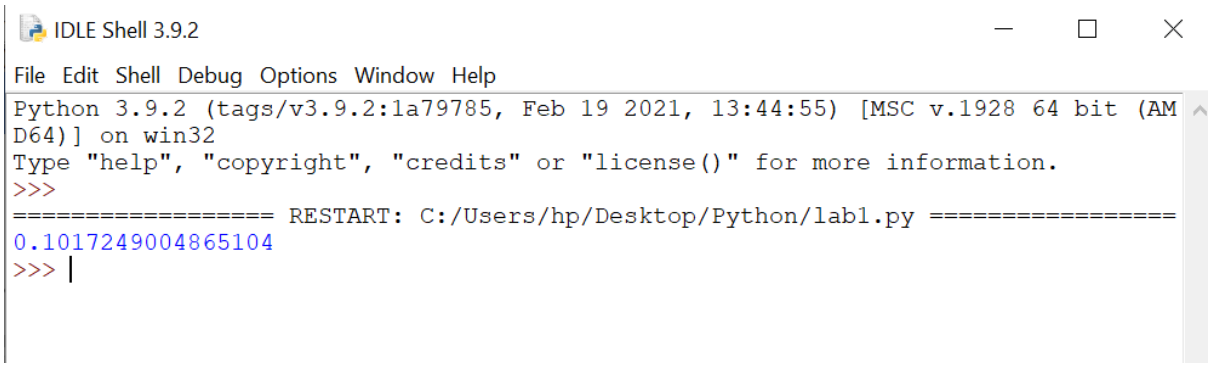
Lab Exercises:

1) Write a program that computes and prints the result of 512 − 282/47 · 48 + 5. It is roughly .1017.

Code

```
print( (512-282)/((47*48)+5) )
```

Output Screenshot



2) Write a program that asks the user for a weight in kilograms and converts it to pounds. There are 2.2 pounds in a kilogram.

Code

```
kg = int(input("Enter the weight in kilograms"))
print(f"The converted weight in pounds is equal to {2.2*kg} pounds")
```

Output Screenshot

```
IDLE Shell 3.9.2
File  Edit  Shell  Debug  Options  Window  Help
Python 3.9.2 (tags/v3.9.2:1a79785, Feb 19 2021, 13:44:55) [MSC v.1928 64 bit (AM
D64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
=============== RESTART: C:\Users\hp\Desktop\Python\Lab1\prog2.py ==============
Enter the weight in kilograms5
The converted weight in pounds is equal to 11.0 pounds
>>>
```

3) Write a program that asks the user to enter three numbers (use three separate input statements). Create variables called total and average that hold the sum and average of the three numbers and print out the values of total and average.

Code

```
x=int(input("Enter first value: "))
y=int(input("Enter second value: "))
z=int(input("Enter third value: "))
total=x+y+z
average = total/3
print(f"Sum of three numbers is {total}")
print(f"Average of three numbers is {average}")
```

Output Screenshot

```
IDLE Shell 3.9.2
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Python 3.9.2 (tags/v3.9.2:1a79785, Feb 19 2021, 13:44:55) [MSC v.1928 64 bit (AM
D64)] on win32
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>>>
=============== RESTART: C:\Users\hp\Desktop\Python\Lab1\prog3.py ==============
Enter first value: 1
Enter second value: 2
Enter third value: 3
Sum of three numbers is 6
Average of three numbers is 2.0
>>>
```

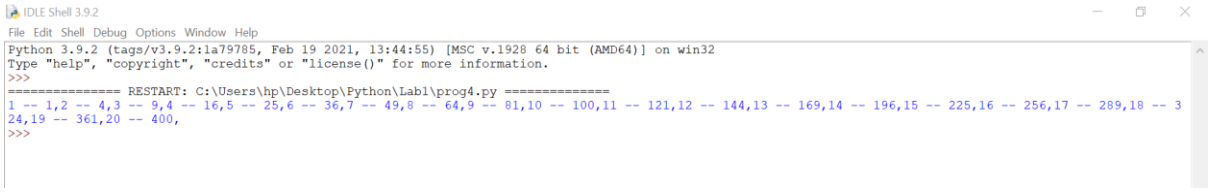
4) Write a program that prints out a list of the integers from 1 to 20 and their squares. The output should look like this:
Output - 1 -- 1, 2 -- 4, 3 -- 9, ... , 200 - 400.

Code

```
s = ""
```

```
for i in range(1,21):
    s = s+f"{i} -- {i*i},"
print(s)
```

Output Screenshot



5) The Fibonacci numbers are the sequence below, where the first two numbers are 1, and each number thereafter is the sum of the two preceding numbers. Write a program that asks the user how many Fibonacci numbers to print and then prints that many. [use while and for loop] 1, 1, 2, 3, 5, 8, 13, 21, 34, 55, 89 . .

Code

```
n = int(input("Enter the number of fibonacci numbers to print"))
f1 = 1
f2 = 1
if n==1:
    print(f1)
elif n==2:
    print(f1)
    print(f2)
elif n>2:
    print(f1)
    print(f2)
    for i in range(2,n):
        f3 = f1 + f2
        print(f3)
        f1 = f2
        f2 = f3
else:
    print("Please enter the valid number")
```

Output Screenshot

```
IDLE Shell 3.9.2
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>>>
============== RESTART: C:\Users\hp\Desktop\Python\Lab1\prog5.py ==============
Enter the number of fibonacci numbers to print6
1
1
2
3
5
8
>>>
```

6) Write a Python program that computes the factorial of a number. The factorial, n!, of a number n is the product of all the integers between 1 and n, including n. For instance, 5! = 1 x 2 x 3 x 4 x 5 = 120. [use while and for loop].

<u>Code</u>

i) Using for loop

```
num = int(input("enter a number: "))

fac = 1

for i in range(1, num + 1):
  fac = fac * i

print("factorial of ", num, " is ", fac)
```

ii) Using while loop :

```
fac = 1
i = 1

while i <= num:
        fac = fac * i
        i = i + 1

print("factorial of ", num, " is ", fac)
```

<u>Output Screenshot</u>

```
IDLE Shell 3.9.2
File  Edit  Shell  Debug  Options  Window  Help
Python 3.9.2 (tags/v3.9.2:1a79785, Feb 19 2021, 13:44:55) [MSC v.1928 64 bit (AM
D64)] on win32
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>>>
============== RESTART: C:\Users\hp\Desktop\Python\Lab1\prog6.py ==============
enter a number: 5
factorial of  5  is  120
>>>
```

7) Write a Python program that asks the user to enter a string. The program should then print the following:
(a) The total number of characters in the string
(b) The string repeated 10 times
(c) The first character of the string (remember that string indices start at 0)
(d) The first three characters of the string
(e) The last three characters of the string
(f) The string backwards
(g) The seventh character of the string if the string is long enough and a message otherwise
(h) The string with its first and last characters removed
(i) The string in all caps
(j) The string with every a replaced with an e
(k) The string with every letter replaced by a space

<u>Code</u>

```
s = input("Enter the string")
total_char = 0
for i in s:
    total_char = total_char + 1
print(f"The total number of characters in the string is {total_char}")
print(f"The string repeated 10 times : {s*10}")
print(f"The first character of the string is {s[0]}")
print(f"The first three characters of the string is {s[:3]}")
print(f"The last three characters of the string is {s[-3:]}")
print(f"The string backwards is {s[::-1]}")
if len(s)<7:
    print("The string is not long enough")
else:
    print(f"The seventh character of the string is {s[6]}")
print(f"The string with its first and last characters removed is {s[1:-1]}")
print(f"The string in all caps is {s.upper()}")
print(f"The string with every 'a' replaced with an 'e' is {s.replace('a','e')}")
for i in range(0,len(s)):
    if s[i].isalpha()==True:
        s=s.replace(s[i],' ')
print(f"The string with every letter replaced by a space is {s}")
```

Output Screenshot

```
IDLE Shell 3.9.2
File  Edit  Shell  Debug  Options  Window  Help
Python 3.9.2 (tags/v3.9.2:1a79785, Feb 19 2021, 13:44:55) [MSC v.1928 64 bit (AMD64)] on win32
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>>>
============== RESTART: C:\Users\hp\Desktop\Python\Lab1\prog7.py ==============
Enter the string10 is even
The total number of characters in the string is 10
The string repeated 10 times : 10 is even10 is even10 is even10 is even10 is even10 is even10 is even10 is even10 is even10 is even
The first character of the string is 1
The first three characters of the string is 10
The last three characters of the string is ven
The string backwards is neve si 01
The seventh character of the string is e
The string with its first and last characters removed is 0 is eve
The string in all caps is 10 IS EVEN
The string with every 'a' replaced with an 'e' is 10 is even
The string with every letter replaced by a space is 10
>>> |
```

8) At a certain school, student email addresses end with @student.college.edu, while professor email addresses end with @prof.college.edu. Write a Python program that first asks the user how many email addresses they will be entering, and then has the user enter those addresses. After all the email addresses are entered, the program should print out a message indicating either that all the addresses are student addresses or that there were some professor addresses entered.

Code

```
n = int(input("Enter the number of email addresses to be entered : "))
print(f"Enter {n} email addresses : ")
email = input()
email_list = email.split(" ")
count = 0
for x in email_list:
    if 'prof' in x:
        count=count+1
if count>0:
    print(f"There are {count} professor email addresses")
else:
    print("All email addresses are student email addresses")
```

Output Screenshot

```
IDLE Shell 3.9.2
File  Edit  Shell  Debug  Options  Window  Help
Python 3.9.2 (tags/v3.9.2:1a79785, Feb 19 2021, 13:44:55) [MSC v.1928 64 bit (AM
D64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
=============== RESTART: C:\Users\hp\Desktop\Python\Lab1\prog8.py ==============
Enter the number of email addresses to be entered : 3
Enter 3 email addresses :
rajat@student.college.edu rajat@prof.college.edu pranam@student.college.edu
There are 1 professor email addresses
>>>
```

9) Start with the list [8,9,10]. Write a Python program to do the following:
(a) Set the second entry (index 1) to 17
(b) Add 4, 5, and 6 to the end of the list
(c) Remove the first entry from the list
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(d) Sort the list
(e) Double the list
(f) Insert 25 at index 3
The final list should equal [4,5,6,25,10,17,4,5,6,10,17]

Code

```
list=[8,9,10]
print("Orignal List: ",list)
list[1]=17
print("List after modifing index 1: ",list)
list.extend([4,5,6])
print("List after appending 4,5 and 6: ",list)
list.pop(0)
print("List after removing first element: ",list)
list.sort()
print("List after sorting:",list)
list=list*2
print("List after doubling:",list)
list.insert(3, 25)
print("After inserting 25:",list)
```

Output Screenshot

```
IDLE Shell 3.9.2
File  Edit  Shell  Debug  Options  Window  Help
Python 3.9.2 (tags/v3.9.2:1a79785, Feb 19 2021, 13:44:55) [MSC v.1928 64 bit (AM
D64)] on win32
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>>>
============== RESTART: C:\Users\hp\Desktop\Python\Lab1\prog9.py ==============
Orignal List:  [8, 9, 10]
List after modifing index 1:  [8, 17, 10]
List after appending 4,5 and 6:  [8, 17, 10, 4, 5, 6]
List after removing first element:  [17, 10, 4, 5, 6]
List after sorting: [4, 5, 6, 10, 17]
List after doubling: [4, 5, 6, 10, 17, 4, 5, 6, 10, 17]
After inserting 25: [4, 5, 6, 25, 10, 17, 4, 5, 6, 10, 17]
>>> |
```

10. Write a Python program that asks the user to enter some text and then counts how many articles are in the text. Articles are the words 'a', 'an', and 'the'.

<u>Code</u>

```
s = input("Enter the string")
chars = s.split(" ")
a = chars.count('a')+chars.count('A')
an = chars.count('an')+chars.count('An')
the = chars.count('the')+chars.count('The')
print(f"The total number of articles in the string is {a+an+the}")
```

<u>Output Screenshot</u>

```
IDLE Shell 3.9.2
File  Edit  Shell  Debug  Options  Window  Help
Python 3.9.2 (tags/v3.9.2:1a79785, Feb 19 2021, 13:44:55) [MSC v.1928 64 bit (AM
D64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
============== RESTART: C:\Users\hp\Desktop\Python\Lab1\prog10.py ==============
Enter the stringAn apple is a fruit that is there in the bucket of a lady.
The total number of articles in the string is 4
>>> |
```